

A102 · Learning Sparse Gaussian Graphical Models from Correlated Data

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Gaussian graphical models (GGMs) are widely used in biomedical research to characterize conditional dependence structures among biological, clinical, and social factors. Contemporary biomedical studies increasingly rely on clustered and longitudinal data that produce correlated observations, and ignoring these correlations can generate networks with false-positive edges. To address this challenge, we recently introduced a cluster-based bootstrap algorithm for learning GGMs from correlated data. While this approach effectively controls false-positive edges, it often produces overly connected networks because partial correlations of any magnitude are retained. Here, we propose a statistical framework that retains only edges with partial correlations exceeding a prespecified threshold. By constructing GGMs over a sequence of thresholds ranging from 0 to 1, we can observe the changes of network structure as weaker edges are progressively removed, allowing stronger connections to reveal biologically meaningful modules. We applied this approach to polygenic risk scores, serum metabolomics, and transcriptomic hallmark aging clocks from the Long Life Family Study. By visualizing networks across different levels of sparsity and evaluating their graphical properties, we demonstrate how variables cluster into coherent modules within biological superclasses while identifying key variables that bridge distinct modules. This dynamic network framework provides an interpretable approach for dissecting the organization of complex biological systems and understanding the relationships among interconnected biological processes.

Learning Sparse Gaussian Graphical Models from Correlated Data

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Abstract

Gaussian graphical models (GGMs) are widely used in biomedical research to characterize conditional dependence structures among biological, clinical, and social factors. Contemporary biomedical studies increasingly rely on clustered and longitudinal data that produce correlated observations, and ignoring these correlations can generate networks with false-positive edges. To address this challenge, we recently introduced a cluster-based bootstrap algorithm for learning GGMs from correlated data. While this approach effectively controls false-positive edges, it often produces overly connected networks because partial correlations of any magnitude are retained. Here, we propose a statistical framework that retains only edges with partial correlations exceeding a prespecified threshold. By constructing GGMs over a sequence of thresholds ranging from 0 to 1, we can observe the changes of network structure as weaker edges are progressively removed, allowing stronger connections to reveal biologically meaningful modules. We applied this approach to polygenic risk scores, serum metabolomics, and transcriptomic hallmark aging clocks from the Long Life Family Study. By visualizing networks across different levels of sparsity and evaluating their graphical properties, we demonstrate how variables cluster into coherent modules within biological superclasses while identifying key variables that bridge distinct modules. This dynamic network framework provides an interpretable approach for dissecting the organization of complex biological systems and understanding the relationships among interconnected biological processes.

Keyword: Gaussian Graphical Models, Correlated Data, Sparse Networks, Omics, Polygenic Risk Scores, Metabolomics, Aging Clocks